Biochemistry Cume (Protein Translation)
January 21, 2010

1. What is the polypeptide sequence specified by the following DNA antisense strand? Assume translation starts after the first initiation codon.
   5′ -TCTGACTATTGAGCTCTCTGGCACATAGCA-3′ **(35 points)**
2. The mRNA specifying the α chain of human hemoglobin contains the base sequence
   . . .UCCAAAUACCGUUAAGCUGGA. . .
   The C-terminal tetrapeptide of the normal α chain, which is specified by part of this sequence, is
   -Ser-Lys-Tyr-Arg.
   In hemoglobin Constant Spring, the corresponding region has the sequence
   -Ser-Lys-Tyr-Arg-Gln-Ala-Gly- . . . **(30 points)**
   Specify the mutation (in the mRNA sequence) that causes hemoglobin Constant Spring.
3. **A**. What is an *amber* codon? **B**. Which amino acids are specified by codons that can be changed to an *amber* codon by a single point mutation? **(35 points)**

Codon Table

First	U	C	A	G	Last
U	Phe	Ser	Tyr	Cys	**U**
	Phe	Ser	Tyr	Cys	**C**
	Leu	Ser	**Stop**	**Stop**	**A**
	Leu	Ser	**Stop**	Trp	**G**
C	Leu	Pro	His	Arg	**U**
	Leu	Pro	His	Arg	**C**
	Leu	Pro	Gln	Arg	**A**
	Leu	Pro	Gln	Arg	**G**
A	Ile	Thr	Asn	Ser	**U**
	Ile	Thr	Asn	Ser	**C**
	Ile	Thr	Lys	Arg	**A**
	Met	Thr	Lys	Arg	**G**
G	Val	Ala	Asp	Gly	**U**
	Val	Ala	Asp	Gly	**C**
	Val	Ala	Glu	Gly	**A**
	Val	Ala	Glu	Gly	**G**